Weibing Zhao (Ivan)

Software Architect · Full-Stack Development · Team Leadership

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PROFESSIONAL SUMMARY

Senior software professional combining deep expertise in software R&D, architecture design, and technical management. Solid engineering foundation and outstanding full-stack capability, backed by years of experience in software development and system architecture, with a strong track record of delivering large-scale projects and leading technical teams.

1. Expert in Node.js/JavaScript/TypeScript full-stack development and proficient in the Java ecosystem. Comprehensive, production-proven technology stack; able to independently own the entire workflow, from requirements breakdown and architecture design through implementation and iterative optimization.
2. Focused on AI Agent architecture and engineering-efficiency innovation. Experienced in integrating AI into real-world development workflows, restructuring and optimizing R&D processes to raise team productivity and delivery speed, with a mature AI-engineering mindset and hands-on delivery experience.
3. Extensive experience architecting user-facing products end to end. Skilled at orchestrating overall architecture planning, test infrastructure, and CI pipeline optimization across the full delivery chain, balancing system stability, scalability, and development efficiency to sustain long-term, high-quality product iteration.
4. Proven leader of mid-size engineering teams. Built and led a core technical team of 12; drove department-level engineering standards, architecture evolution, technical reviews, and key technical decisions, establishing a standardized, closed-loop technical governance process that reduced internal friction and increased team output.
5. Strong engineering depth combined with sharp business acumen, spanning technical excellence, team management, and innovation delivery. Honored as a Nokia Bell Labs DMTS (Distinguished Member of Technical Staff) and Business Line Model Worker; won First Prize in the R&D Center's AI Application Competition.

WORK EXPERIENCE

Nokia Solutions and Networks Technology (Beijing) Co., Ltd. — Software Architect Oct 2011 – Present

- As the department's technical lead, drove technical iteration and architecture evolution of core products; led the C/S-to-B/S migration of the management software, cutting O&M costs and enabling remote access.
- Built a shared architecture reused across 4G and 5G products, raising code reuse by 70%.
- Built and led a 12-person technical management team; established architecture-review and technical decision-making mechanisms covering all product lines, standardizing technology selection and evolution, and achieving closed-loop technical governance for the department.
- Established quality and performance governance — automated testing plus performance-baseline regression — to safeguard delivery quality and operational stability.
- Partnered with product managers and customers to clarify requirements and design technical solutions.

Career Progression: Software Engineer (Oct 2011 – May 2015) → Tech Lead (May 2015 – Jul 2018) → Software Architect (Jul 2018 – Present)

Tata Consultancy Services (TCS) — Software Development Engineer Sep 2008 – Aug 2011

Hangzhou (Sep 2008 – Sep 2009) / On-site in Singapore (Sep 2009 – Aug 2011)

- Iterated features for Motorola's mobile-phone distribution-channel system; developed and maintained the OnePA and ERAS government systems for the Singapore People's Association.
- Independently owned the full workflow, from requirements analysis and technical architecture through implementation, integration testing, UAT, and delivery.
- Delivered multiple third-party integration solutions; won the Star of the Month Award several times.

PROJECT EXPERIENCE

WebEM — Web-based Network Element Management System · Software Architect

Jul 2018 – Present

Description: A next-generation web-based element management system for 4G/5G radio access networks, enabling remote browser access to gNB/eNB network elements and full-lifecycle base-station management.

- Led architecture design and core implementation; abstracted a unified backend so one set of features served multiple frontends — 70% module reuse and markedly lower cross-team coordination costs.
- Unified code across CLI and web clients; implemented API versioning with backward compatibility and a consistent user experience.
- Established quality and performance governance (automated testing, performance baselines, regression mechanisms), reaching 95% automated-test coverage, cutting the defect rate by 90%, and improving system performance by 50%.

Tech Stack: JavaScript/TypeScript, Angular, React, WebAssembly (Rust), Node.js, Puppeteer, Cucumber, Protobuf, C++, Python, Electron

AI-Empowered R&D Efficiency · Technical Manager

Feb 2024 – Apr 2026

Description: Applied state-of-the-art AI and Coding Agents across the full development workflow to boost team productivity.

- Built a RAG knowledge-base system indexing project documents and team know-how; exposed MCP interfaces and customized project Skills.
- Planned and led a department-level AI upskilling program, delivering systematic training to 70+ engineers.
- Improved R&D efficiency by 40%.

Tech Stack: Milvus, MySQL, MinIO, Redis, Node.js, NestJS, React, Vite

PDL — Parameter Logic Validation Language · Tech Lead

Oct 2016 – Jul 2018

Description: A domain-specific language (DSL) for describing relationships between parameters, enabling non-programmer requirements staff to implement parameter validation.

- Led the full lifecycle development of the PDL product; coordinated cross-team collaboration, unifying requirements management and interface specifications for efficient multi-team delivery.
- Designed and brought online a complete toolchain for editing, compiling, and testing PDL rules — covering development through verification and improving rule-development efficiency by 80%.
- Designed configuration-file upgrade/migration and PDL integration tests, reducing the defect rate by 75%.

Tech Stack: Java, JavaScript, Python, PostgreSQL, C++

FZC — Small Cell Cluster Management System · Tech Lead

May 2015 – Oct 2016

Description: A centralized monitoring and management system for small-cell clusters, collecting and visualizing base-station operational data in real time.

- As tech lead, interfaced with the US team on business requirements and owned the overall architecture design and feature breakdown.
- Coordinated cross-functional teams (frontend, backend, QA, product); set development plans and controlled schedules to ensure on-time delivery.

Tech Stack: Java, Swing, TCP/IP

BTSSM — Desktop Network Element Management System · Software Engineer Oct 2011 – May 2015

Description: A desktop system for centralized base-station monitoring and management, collecting and visualizing operational data in real time.

- Designed the multi-base-station management UI, achieving WYSIWYG configuration without compromising performance.
- Designed and iterated core modules; planned and drove delivery — 100% on-time delivery of core-module requirements.
- Championed Clean Code practices and JUnit unit testing; organized Developer Group Meetings to codify engineering best practices.
- Resolved multiple system-level critical issues, cutting mean time to repair (MTTR) by 50%.

Tech Stack: Java, Swing, TCP/IP

Singapore People's Association Government Information System · Software Development Engineer Sep 2009 – Aug 2011

Description: A digital community-governance platform for the Singapore People's Association, integrating community-event management, volunteer coordination, and citizen-feedback services.

- Developed new features for the OnePA and ERAS systems, independently owning the full cycle: requirements analysis, estimation, architecture, implementation, integration testing, UAT, and delivery.

Tech Stack: J2EE, Swing, Tomcat, Struts, Hibernate, iBatis, Oracle, MySQL

EDUCATION

Xiangtan University (National "Double First-Class" University) Sep 2004 – Jul 2008

Information Management and Information Systems · Bachelor's Degree

HONORS & AWARDS

- **2026** First Prize, AI Application Competition, Hangzhou R&D Center
- **2026** Model Worker, Business Line
- **2025** UI Department Best Practice Award — "For Empowered"
- **2025** MN China Business Award — People Excellence
- **2024** UI Department Best Practice Award — "For Empowered"
- **2022** Outstanding Achievement for Unify WebEM CoDe
- **2022** MN China 2nd LanDing Award — Phoenix Quality
- **2020** Nokia Bell Labs DMTS — Distinguished Member of Technical Staff
- **2018** DU R&D Achievement Award — WebEM Delivery with High Quality
- **2017** HZ TC Recognition Award — Harmonized Object Models for SRAN and LTE Project
- **2017** HZ TC Recognition Award — BTSSM FL17 Taskforce Team
- **2017** Top BTSSM Pronto Killer
- **2016** Nokia HZ Technology Center "Meet Again Debate Contest" Winner Team Award
- **2016** Department Q3 "Best Trainer"
- **Multi** Star of the Month Award (multiple times)

CERTIFICATES

- China Computer Technology and Software Professional Qualification (Soft Exam) — Network Engineer
- China Computer Technology and Software Professional Qualification (Soft Exam) — Programmer
- National Computer Rank Examination (NCRE) — Level 3
- College English Test — Band 4 (CET-4)